Distributional chaos is generic among uniformly convergent non-autonomous systems on the interval

Yitzchak Shmalo
Einstein Institute of Mathematics, The Hebrew University of Jerusalem, Jerusalem, Israel
yitzchak.shmalo@gmail.com

June 2026

Abstract

Balibrea and Rucká [*Genericity of distributional chaos in non-autonomous systems*, arXiv:2410.10288] proved that in the space of convergent non-autonomous systems of surjective interval maps, the systems with distributional chaos of type 1 (DC1) form a dense set that is neither open nor closed, and stated explicitly that the residuality of this set remains an open problem. We solve the problem in the canonical complete setting: with the uniform-in-time supremum metric, the DC1 systems contain an *open dense* subset of the space of convergent systems — namely, the systems whose limit map carries a strict one-step horseshoe — and are therefore residual. Thus DC1 is a generic property of uniformly convergent non-autonomous systems on the interval, in contrast with the Cantor set, where the same authors proved genericity of the absence of distributional chaos. The proof combines a quantitative openness–density analysis of strict horseshoes for the limit map with the inheritance theorem of Štefánková and the Schweizer–Smítal characterization of positive entropy on the interval. Two complements are included: infinite topological entropy is likewise generic in this space, by a self-contained common-horseshoe estimate, which removes the equicontinuity hypothesis from the genericity theorem of Balibrea, Smítal and Štefánková (we observe that uniform convergence already implies equicontinuity of the system); and we delineate precisely what remains open for the integral-type metrics that distinguish pointwise from uniform convergence, where the ambient space fails to be complete.

1 Introduction

A non-autonomous discrete dynamical system (nads) on the compact interval $I = [0,1]$ is a sequence $f_{1,\infty} = (f_n)_{n\geq 1}$ of continuous surjective maps $f_n : I \to I$, acting through the compositions $f^n_{1,\infty} = f_n \circ \cdots \circ f_1$. Following [1, 2], let $C_s(I)$ denote the continuous surjections of I with the supremum metric $\rho_{\sup}(f,g) = \sup_x |f(x) - g(x)|$, let

$$\mathcal{F}_p(I) \;:=\; \Big\{(f_n)_{n\geq 1} \in C_s(I)^{\mathbb{N}} :\; (f_n) \text{ converges in } \rho_{\sup}\Big\}, \qquad \rho_{\mathcal{F}}\big(f_{1,\infty}, g_{1,\infty}\big) := \sup_{n\geq 1} \rho_{\sup}(f_n, g_n),$$

and write $\lambda(f_{1,\infty}) := \lim_n f_n \in C(I)$ for the limit map. (As observed in [1], with supremum metrics every convergent system converges uniformly, so $\mathcal{F}_p(I)$ is the space of *uniformly* convergent systems; we return to other metrics in Section 5.) A pair $x \neq y$ is DC1-scrambled for $f_{1,\infty}$ if its upper distributional function is identically 1 and its lower distributional function

vanishes at some $t_0 > 0$ (Section 2); the system is *DC1* if it has an uncountable set of pairwise DC1-scrambled points. Let $\mathcal{F}_{DC1}(I) \subseteq \mathcal{F}_p(I)$ denote the DC1 systems.

Balibrea and Rucká [1] proved that $\mathcal{F}_{DC1}(I)$ is dense in $\mathcal{F}_p(I)$ (for every admissible metric), that on the Cantor set the *complement* of distributional chaos is generic, and that on the interval $\mathcal{F}_{DC1}(I)$ is neither open nor closed — and they singled out the remaining question ([1, Section 4]):

> "Note that Lemma 2 does not say $\mathcal{F}_{DC1}(I)$ is not residual. This problem stays open."

This paper resolves the problem in the canonical complete setting.

Theorem 1.1. *The set $\mathcal{F}_{DC1}(I)$ of DC1 systems contains an open dense subset of $(\mathcal{F}_p(I), \rho_{\mathcal{F}})$, namely*

$$\lambda^{-1}(\mathcal{O}) = \{ f_{1,\infty} \in \mathcal{F}_p(I) : \ \lambda(f_{1,\infty}) \in \mathcal{O} \},$$

where $\mathcal{O} \subseteq C_s(I)$ is the (open, dense) set of surjective maps carrying a strict one-step horseshoe (Definition 3.1). In particular $\mathcal{F}_{DC1}(I)$ is residual: DC1 is a generic property of uniformly convergent non-autonomous systems on the interval.

The point is that the failure of openness established in [1, Lemma 2] is a boundary phenomenon: a DC1 system may converge to a non-chaotic limit, and such systems can be tail-perturbed out of $\mathcal{F}_{DC1}(I)$. But systems whose *limit* is robustly chaotic form an open set — openness of the horseshoe is measured for the limit map, where the metric $\rho_{\mathcal{F}}$ controls $\rho_{\sup}$-perturbations of the limit — and the inheritance theorem of Štefánková [7] pushes the chaos of the limit back onto every uniformly convergent system above it. Residual sets in a complete metric space may well be neither open nor closed; the open dense *subset* is the correct object, and this observation is the entire content of the solution.

The same mechanism yields a second genericity statement, this time self-contained.

Theorem 1.2. *The systems with infinite topological entropy contain a dense G_δ subset of $(\mathcal{F}_p(I), \rho_{\mathcal{F}})$; indeed for every $m \geq 2$ the set of systems whose limit map carries a strict m-branch horseshoe is open and dense, and any such system satisfies $h(f_{1,\infty}) \geq \log m$. Hence infinite topological entropy is generic in $\mathcal{F}_p(I)$.*

Theorem 1.2 recovers the genericity theorem of Balibrea, Smítal and Štefánková [2, Theorem A-type result], which was stated for *uniformly convergent and equicontinuous* systems, and removes the second hypothesis — vacuously, as it turns out:

Proposition 1.3. *Every uniformly convergent sequence (f_n) of continuous self-maps of a compact metric space is an equicontinuous family. Consequently "uniformly convergent and equicontinuous" defines the same class of systems as "uniformly convergent".*

Finally, Section 5 discusses the metrics of class ϱ_1 of [1] (such as the integral metric), which genuinely distinguish pointwise from uniform convergence. There the ambient space fails to be complete, the Baire-category formulation of genericity requires a choice of completion or framework, and the entropy and chaos transfer between system and limit breaks down ([1, Section 3]); the genericity problem in that setting remains open, and we state it precisely.

2 Preliminaries

Throughout, $I=[0,1]$ with the euclidean metric d. For $f_{1,\infty}=(f_n)\in C_s(I)^{\mathbb{N}}$ put $f^0_{1,\infty}=\mathrm{id}$ and $f^n_{1,\infty}=f_n\circ\cdots\circ f_1$. For $x,y\in I$ the lower and upper distributional functions are

$$\psi_{xy}(t)=\liminf_{n\to\infty}\frac{1}{n}\#\big\{0\le j<n:\ d(f^j_{1,\infty}(x),f^j_{1,\infty}(y))<t\big\},\qquad \psi^*_{xy}(t)=\limsup_{n\to\infty}(\cdots),$$

and (x,y) is a DC1-scrambled pair if $\psi^*_{xy}\equiv 1$ on $(0,\infty)$ and $\psi_{xy}(t_0)=0$ for some $t_0>0$. The system is DC1 if some uncountable $S\subseteq I$ consists of pairwise DC1-scrambled points. Topological entropy $h(f_{1,\infty})$ is defined through (n,ε)-separated sets as in [5, 1]; for an autonomous map f, $h(f)$ is the usual entropy and we identify f with the constant system $(f,f,\dots)$.

We use three known results.

Theorem 2.1 (Schweizer–Smítal [6])**.** *A continuous interval map has positive topological entropy if and only if it is distributionally chaotic; in particular $h(f)>0$ implies that f is DC1.*

Theorem 2.2 (Štefánková [7])**.** *Let $(f_n)\subset C_s(I)$ converge uniformly to f. If f is DC1, then the non-autonomous system $f_{1,\infty}$ is DC1.*

Lemma 2.3. *$(\mathcal{F}_p(I),\rho_{\mathcal{F}})$ is a complete metric space; hence it is a Baire space, and residuality in $\mathcal{F}_p(I)$ is meaningful.*

Proof. $C_s(I)$ is complete: a $\rho_{\sup}$-Cauchy sequence of continuous surjections converges uniformly to a continuous map, which is surjective because a uniform limit of surjective self-maps of a compact space is surjective (if $u_k\to u$ and $u_k(x_k)=y$ with $x_k\to x$ along a subsequence, then $u(x)=y$). Let $(f^{(k)}_{1,\infty})_k$ be $\rho_{\mathcal{F}}$-Cauchy in $\mathcal{F}_p(I)$. For each n the maps $f^{(k)}_n$ form a $\rho_{\sup}$-Cauchy sequence with modulus independent of n, hence converge uniformly to some $g_n\in C_s(I)$, and $\rho_{\mathcal{F}}(f^{(k)}_{1,\infty},g_{1,\infty})\to 0$. It remains to check $g_{1,\infty}\in\mathcal{F}_p(I)$, i.e. $(g_n)_n$ is $\rho_{\sup}$-Cauchy. Given $\varepsilon>0$ pick k with $\rho_{\mathcal{F}}(f^{(k)},g)<\varepsilon/3$ and then N with $\rho_{\sup}(f^{(k)}_n,f^{(k)}_m)<\varepsilon/3$ for $n,m\ge N$ (possible since $(f^{(k)}_n)_n$ converges). Then $\rho_{\sup}(g_n,g_m)<\varepsilon$ for $n,m\ge N$. □

Proof of Proposition 1.3. Let $f_n\to f$ uniformly on the compact space X and let $\varepsilon>0$. Choose N with $\rho_{\sup}(f_n,f)<\varepsilon/3$ for $n>N$. The maps $f,f_1,\dots,f_N$ are finitely many continuous maps on a compact space, hence uniformly continuous with a common modulus: there is $\delta>0$ such that $d(x,y)<\delta$ implies $d(g(x),g(y))<\varepsilon/3$ for $g\in\{f,f_1,\dots,f_N\}$. For $n>N$ and $d(x,y)<\delta$,

$$d(f_n(x),f_n(y))\le d(f_n(x),f(x))+d(f(x),f(y))+d(f(y),f_n(y))<\varepsilon.$$

Thus the family (f_n) is equicontinuous. □

3 Strict horseshoes for the limit map

Definition 3.1. *Let $m\ge 2$. A map $g\in C(I)$ carries a* strict m-branch horseshoe *with margin $\theta>0$ if there are pairwise disjoint nonempty closed intervals $A_1<A_2<\cdots<A_m$ in I such that, with $[u,v]:=[\min A_1,\max A_m]$ (the convex hull of $\bigcup_i A_i$),*

$$g(A_i)\ \supseteq\ [u-\theta,\,v+\theta]\cap I\quad\text{and}\quad [u-\theta,v+\theta]\subseteq(0,1)\qquad(1\le i\le m).$$

Let $\mathcal{O}_m\subseteq C_s(I)$ be the set of surjective maps carrying a strict m-branch horseshoe (for some intervals and margin), and $\mathcal{O}:=\mathcal{O}_2$.

Lemma 3.2 (Openness)**.** *For every $m \geq 2$, $\mathcal{O}_m$ is open in $(C_s(I), \rho_{\sup})$: if g carries a strict m-horseshoe with margin θ and $\rho_{\sup}(h, g) < \theta/2$, then h carries the same intervals as a strict m-horseshoe with margin $\theta/2$.*

Proof. $g(A_i)$ is an interval containing $[u - \theta, v + \theta]$, so $\min_{A_i} g \leq u - \theta$ and $\max_{A_i} g \geq v + \theta$. If $\rho_{\sup}(h, g) < \theta/2$ then $\min_{A_i} h \leq u - \theta/2$ and $\max_{A_i} h \geq v + \theta/2$, and $h(A_i)$, being an interval, contains $[u - \theta/2, v + \theta/2]$. □

Lemma 3.3 (Density)**.** *For every $m \geq 2$, $\mathcal{O}_m$ is dense in $(C_s(I), \rho_{\sup})$.*

Proof. Let $g \in C_s(I)$ and $\varepsilon > 0$. Since g is a continuous surjection of I, it has a fixed point z: the map $x \mapsto g(x) - x$ is ≥ 0 at a preimage of 1 and ≤ 0 at a preimage of 0. Replacing z by a nearby point if necessary we may assume $z \in (0, 1)$ (if the only fixed points are 0 or 1, say $z = 0$, pick $z' \in (0, 1)$ small; then $|g(z') - z'| \leq |g(z') - g(0)| + z' \leq \omega_g(z') + z'$ is small, and the construction below around z' works verbatim with this additional error). We give the construction for a fixed point $z \in (0, 1)$; the general case only changes ε-bookkeeping by the displayed error.

Let ω_g be the modulus of continuity of g, let $R := \max\{\eta, \omega_g(\eta)\} + \eta$, and choose $\eta > 0$ so small that $J := [z - \eta, z + \eta] \subseteq (0, 1)$ and

$$\varepsilon' := 2R = 2\big(\max\{\eta, \omega_g(\eta)\} + \eta\big) < \varepsilon,$$

which is possible since $R \to 0$ as $\eta \to 0$. Let $K := [z - R,\, z + R] \cap I$; note $g(J) \subseteq [z - \omega_g(\eta), z + \omega_g(\eta)] \subseteq K$, since $|g(x) - z| = |g(x) - g(z)| \leq \omega_g(\eta)$ on J. Define $h \in C(I)$ by $h := g$ off J, and on J let h be the piecewise linear map which fixes the endpoints, $h(z \pm \eta) = g(z \pm \eta)$, and zigzags $2m$ times across the interval K: concretely, partition J into $2m + 2$ subintervals; on the first and last, interpolate linearly from $g(z \mp \eta)$ to the zigzag; on the middle $2m$ subintervals let h alternate linearly between the values $\min K$ and $\max K$.

Then h is continuous, and for $x \in J$ both $h(x)$ and $g(x)$ lie in K, so

$$\rho_{\sup}(h, g) \leq \operatorname{diam} K \leq 2R = \varepsilon' < \varepsilon.$$

h is surjective: $h(I) \supseteq g(I \setminus J) \cup h(J) = g(I \setminus J) \cup K \supseteq g(I \setminus J) \cup g(J) = g(I) = I$, because $g(J) \subseteq K = h(J)$.

Finally $h \in \mathcal{O}_m$: the m increasing zigzag branches contain pairwise disjoint closed intervals $A_1 < \cdots < A_m$ in the interior of J with $h(A_i) = K$ for each i. Their convex hull $[u, v]$ satisfies $[u, v] \subseteq J = [z - \eta, z + \eta]$, while K contains $[z - R, z + R] \cap I$ and $R - \eta = \max\{\eta, \omega_g(\eta)\} \geq \eta > 0$. Shrink the intervals A_i slightly so that $[u, v]$ lies in the interior of J, and choose

$$\theta := \tfrac{1}{2} \min\big\{R - \eta,\ \operatorname{dist}(J, \{0, 1\})\big\} > 0.$$

Then $[u - \theta, v + \theta] \subseteq (0, 1)$ and $h(A_i) = K \supseteq [z - R, z + R] \cap I \supseteq [u - \theta, v + \theta]$, since every point of $[u - \theta, v + \theta]$ is within $\eta + \theta \leq R$ of z and lies in $(0, 1)$. Thus h carries a strict m-branch horseshoe with margin θ. □

Remark 3.4. *The proof shows more: the horseshoe may be implanted at any prescribed scale η, and the margin obtained is at least η. For $m = 2$ this is the classical fact that "robust horseshoes are C^0-dense"; we could not locate a reference in the surjective category with explicit margins, so we included the proof. Compare Kawaguchi [4], who proved C^0-density of DC1 maps.*

4 Proofs of the main theorems

Lemma 4.1. *The limit-map function $\lambda : (\mathcal{F}_p(I), \rho_{\mathcal{F}}) \to (C_s(I), \rho_{\sup})$ is 1-Lipschitz.*

Proof. $\rho_{\sup}\big(\lambda(f_{1,\infty}), \lambda(g_{1,\infty})\big) = \lim_n \rho_{\sup}(f_n, g_n) \le \rho_{\mathcal{F}}(f_{1,\infty}, g_{1,\infty})$. □

Lemma 4.2. *For every $m \ge 2$, the set $\lambda^{-1}(\mathcal{O}_m)$ is open and dense in $(\mathcal{F}_p(I), \rho_{\mathcal{F}})$.*

Proof. Openness is immediate from Lemmas 3.2 and 4.1. For density, let $f_{1,\infty} \in \mathcal{F}_p(I)$ with limit f and let $\varepsilon > 0$. By Lemma 3.3 pick $g \in \mathcal{O}_m$ with $\rho_{\sup}(g, f) < \varepsilon/2$, and pick N with $\rho_{\sup}(f_n, f) < \varepsilon/2$ for $n > N$. Define $g_{1,\infty}$ by $g_n := f_n$ for $n \le N$ and $g_n := g$ for $n > N$. Then $g_{1,\infty} \in \mathcal{F}_p(I)$ (its tail is constant), $\lambda(g_{1,\infty}) = g \in \mathcal{O}_m$, and

$$\rho_{\mathcal{F}}(f_{1,\infty}, g_{1,\infty}) = \sup_{n>N} \rho_{\sup}(f_n, g) \le \sup_{n>N} \big[\rho_{\sup}(f_n, f) + \rho_{\sup}(f, g)\big] < \varepsilon.$$

□

Lemma 4.3 (Common horseshoe forces entropy)**.** *Let $f_{1,\infty} \in C_s(I)^{\mathbb{N}}$ and suppose there are $m \ge 2$, $N \ge 0$, pairwise disjoint closed intervals $A_1 < \cdots < A_m$ with convex hull $[u, v]$, and $\theta > 0$ such that*

$$f_n(A_i) \supseteq [u, v] \qquad \text{for all } n > N,\ 1 \le i \le m.$$

Then $h(f_{1,\infty}) \ge \log m$.

Proof. Let $\gamma > 0$ be the minimal gap between consecutive intervals A_i. Fix a horizon $L \ge 1$ and a word $\omega \in \{1, \dots, m\}^L$. Choose $y_L \in A_{\omega_L}$ arbitrarily. Descending from $j = L - 1$ to $j = 1$: given $y_{j+1} \in A_{\omega_{j+1}} \subseteq [u, v]$, the covering hypothesis (applied to $n = N + j > N$) gives $f_{N+j}(A_{\omega_j}) \supseteq [u, v] \ni y_{j+1}$, and since the continuous image of an interval is an interval there is $y_j \in A_{\omega_j}$ with $f_{N+j}(y_j) = y_{j+1}$. This produces $y_1 \in A_{\omega_1}$ whose forward orbit under the tail $(f_{N+1}, f_{N+2}, \dots)$ satisfies $f_{N+j-1} \circ \cdots \circ f_{N+1}(y_1) = y_j \in A_{\omega_j}$ for $j = 1, \dots, L$. Since each f_n is surjective, so is $f^N_{1,\infty}$; choose x_ω with $f^N_{1,\infty}(x_\omega) = y_1$. Then $f^{N+j-1}_{1,\infty}(x_\omega) = y_j \in A_{\omega_j}$ for $j = 1, \dots, L$.

If $\omega \ne \omega'$ differ at position j, the orbits of $x_\omega, x_{\omega'}$ at time $N + j - 1$ lie in distinct intervals $A_{\omega_j} \ne A_{\omega'_j}$, hence are γ-separated. Thus $\{x_\omega\}$ is an $(N + L, \gamma)$-separated set of cardinality m^L, and

$$h(f_{1,\infty}) \ \ge\ \limsup_{L\to\infty} \frac{\log m^L}{N + L} \ =\ \log m.$$

□

Proof of Theorem 1.1. By Lemma 4.2 (with $m = 2$), $\lambda^{-1}(\mathcal{O})$ is open and dense in $\mathcal{F}_p(I)$. Let $f_{1,\infty} \in \lambda^{-1}(\mathcal{O})$ with limit f. By Lemma 4.3 applied to the constant system $(f, f, \dots)$ — the covering hypothesis holds with $N = 0$ since $f(A_i) \supseteq [u - \theta, v + \theta] \supseteq [u, v]$ — we get $h(f) \ge \log 2 > 0$. By Theorem 2.1, f is DC1. By Theorem 2.2, the uniformly convergent system $f_{1,\infty}$ is DC1. Hence $\lambda^{-1}(\mathcal{O}) \subseteq \mathcal{F}_{DC1}(I)$, and $\mathcal{F}_{DC1}(I)$, containing an open dense set, is residual in the complete space $(\mathcal{F}_p(I), \rho_{\mathcal{F}})$ (Lemma 2.3). □

Proof of Theorem 1.2. Fix $m \ge 2$ and let $f_{1,\infty} \in \lambda^{-1}(\mathcal{O}_m)$, with limit f carrying a strict m-horseshoe on intervals $A_1 < \cdots < A_m$ with margin θ. Choose N with $\rho_{\sup}(f_n, f) < \theta/2$ for $n >$

N. By the proof of Lemma 3.2, every f_n with $n > N$ satisfies $f_n(A_i) \supseteq [u-\theta/2, v+\theta/2] \supseteq [u, v]$. By Lemma 4.3, $h(f_{1,\infty}) \geq \log m$. Hence

$$\bigcap_{m\geq 2} \lambda^{-1}(\mathcal{O}_m) \subseteq \{f_{1,\infty} : \ h(f_{1,\infty}) = \infty\},$$

and the left-hand side is a countable intersection of open dense sets (Lemma 4.2), hence a dense G_δ. □

Remark 4.4. *Theorem 1.2 together with Proposition 1.3 subsumes the genericity of infinite entropy for "uniformly convergent and equicontinuous" systems from [2]: by Proposition 1.3 the equicontinuity hypothesis is implied by uniform convergence, and the proof above is independent of [2]. Note also that all three results (Theorems 1.1, 1.2 and the density theorem of [1]) now align with the autonomous interval theory: positive entropy, DC1, and their genericity transfer along the limit map.*

Remark 4.5 (Contrast with the Cantor set)**.** *On the Cantor set, Balibrea and Rucká proved that the* absence *of distributional chaos is generic for convergent systems [1, Lemma 1], following Darji–D'Aniello [3]. Theorem 1.1 shows the interval behaves in the opposite way, completing the table of [1, Section 4]: DC1 is generic in $\mathcal{F}_p(I)(I)$ and non-DC1 is generic in $\mathcal{F}_p(I)(Q)$. The mechanism of the difference is visible in the proofs: on the interval, surjectivity plus the intermediate value theorem make strict horseshoes of the limit map both robust and densely implantable, while on the Cantor set the clopen structure makes adding-machine-like (chaos-free) limits robust and dense.*

5 The metrics distinguishing pointwise from uniform convergence

Balibrea and Rucká formulate their results for arbitrary metrics on $C_s(I)$, split into the class ϱ_2 (metrics, like $\rho_{\sup}$, for which convergence of maps forces uniform convergence) and the class ϱ_1 (metrics, like the integral metric $\rho_{\text{int}}(f, g) = \int_0^1 d(f(x), g(x))\, dx$, for which genuinely pointwise-but-not-uniformly convergent sequences of maps exist) [1, Remark 1]. Theorems 1.1 and 1.2 settle the genericity problem for the supremum metric, the canonical representative of ϱ_2. For other metrics in ϱ_2 the underlying set of systems is the same — every ϱ_2-convergent system is uniformly convergent — but residuality depends on the topology of the system space, so each such metric for which the space is Baire requires checking that the horseshoe class $\lambda^{-1}(\mathcal{O})$ remains open and dense; for metrics uniformly equivalent to $\rho_{\sup}$ this is automatic.

For ϱ_1-metrics the situation is genuinely different, for three reasons already implicit in [1]. First, $(C_s(I), \rho_{\text{int}})$ is not complete — pointwise limits of continuous surjections need not be continuous — so the system space is not complete either, and [1] must explicitly demand continuity of limit maps; the Baire category theorem is then unavailable without either passing to a completion (changing the space) or verifying the Baire property by hand. Second, the entropy bridge fails: there are systems with positive (even infinite) entropy converging pointwise to the identity [1, Figure 1], so the limit map no longer certifies the dynamics of the system, and our key mechanism — openness of chaos *at the level of the limit* — has no analogue. Third, the inheritance theorem (Theorem 2.2) is a theorem about uniform convergence. We therefore record what remains of the original question:

Open problem. Let $\rho \in \varrho_1$ (for instance $\rho = \rho_{\mathrm{int}}$), and consider the set F of pointwise convergent systems of surjective interval maps with continuous limit, with the metric $\sup_n \rho(f_n, g_n)$. Is the subspace F Baire? If so, is DC1 generic in it?

Our results show that any positive answer must use a mechanism that does not factor through the limit map; the examples of [1, Section 3] (positive entropy, eventually-fixed-point systems converging pointwise to the identity) suggest that smallness at the level of the limit and largeness at the level of the system can coexist on residual-looking sets, and we expect the answer to depend on the metric within ϱ_1.

Acknowledgments

The research presented in this paper was supported by the European Research Council (ERC) under the European Union's Horizon Europe research and innovation programme (grant agreement No. 101041711), by the Israel Science Foundation (grant number 2258/19) and by the Israel Science Foundation (ISF Grant 4101/25).

Declaration on generative AI

During the preparation of this work the author used generative AI tools to accelerate drafting and revision. All mathematical claims, proofs, and bibliographic information were subsequently reviewed and edited by the author, who takes full responsibility for the content of the manuscript.

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